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The role of play in developing sensory-motor coordination in the autistic child "picture matching game as a model"

دور اللعب في تنمية التآزر الحسي الحركي عند الطفل التوحدي " لعبة تطابق الصور نموذجا "

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Abstract:

It is acknowledged that play has a role in the development of sensory-motor synergy in autistic children. This study aims to highlight this role through a picture-matching game as a model. A case study approach was adopted for an autistic child from the Psycho-Pedagogical Centre Grace - Camp, using an interview, observation and a programme based on a picture matching game to develop sensory-motor synergy. The results indicated that the programme was effective in the development of fine motor movements of the autistic child. Consequently, it can be concluded that play has a role in the development of sensory-motor synergy in autistic children. The objective of this study is to reveal the role of using play as a therapeutic technique for the sensory-motor aspect in autistic children and to identify the importance and effectiveness of the picture matching game as a therapeutic technique for the sensory-motor aspect in autistic children.

Keywords: play, sensorimotor synergy, autistic child, picture matching game.

الملخص:

المتعارف عليه أن للعب دور في تنمية التآزر الحسي الحركي عند الطفل التوحدي، ومن هذا المنطلق، تتبنى هذه الدراسة إبراز هذا الدور من خلال لعبة تطابق الصور نموذجا. حيث اعتمدنا على منهج دراسة الحالة لطفل توحدي من المركز النفسي البيداغوجي غريس – معسكر باستخدام المقابلة والملاحظة وبرنامج قائم على لعبة تطابق الصور لتنمية التآزر الحسي الحركي. وقد خلصت النتائج الى وجود فعالية للبرنامج في تنمية الحركات الدقيقة للطفل التوحدي، ومنه فإن للعب دور في تنمية التآزر الحسي الحركي عند الطفل التوحدي. وتهدف الدراسة الكشف عن دور استخدام اللعب كتقنية علاجية للجانب الحسي الحركي عند الطفل التوحدي، التعرف على أهمية وفعالية لعبة تطابق الصور كتقنية علاجية للجانب الحسي الحركي عند الطفل التوحدي.

الكلمات المفتاحية : اللعب، التآزر الحسي الحركي، الطفل التوحدي، لعبة تطابق الصور.

1. INTRODUCTION

Play is a significant aspect of childhood, and among the most crucial and vital needs of the child, it is imperative that we pay attention to and care for. As we find evidence of children imitating others in various activities since the beginning of history, it is evident that play is an important aspect of childhood. The activities of sweeping and washing for females and driving cars and motorcycles for males, and other behaviours, motivated us to understand the desires and tendencies of this child. Play is an important characteristic and feature in childhood, and a means of discovery, fun and pleasure. It is also a motive for maturity and self-formation, and the acquisition of a set of skills, including physical, linguistic, and emotional. Emotional and even the sensory-motor aspect, the latter of which is difficult for the autistic child. Play is an important feature and characteristic in childhood, a means of discovery, fun and pleasure, a motivation for maturity, self-formation and the acquisition of a set of skills, including physical, linguistic, emotional and even the sensory-motor aspect, the latter of which the autistic child finds difficult. In such cases, play has been used as a therapeutic technique in order to develop it and help him develop his fine

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muscles and use them properly. Among these games, we find the picture matching game, which serves as a model for that.

Play is considered an important, necessary and pivotal activity for humans in general and children in particular. It is the main role of play within the child's life, helping physical, mental and even emotional development. This technique enables the child to perform many diverse skills, prepare for learning and promote the integration between their mental development and social life. Play becomes a language with which the child communicates and games become words. [1, p. 2]

2. HYPOTHESES OF THE STUDY

The study was based on the hypothesis that Picture matching game has a role in the development of sensory-motor synergy in autistic children.

3. DEFINING CONCEPTS

This study includes a set of terms.

3.1. Play

A free activity carried out by the child to achieve pleasure, fun and happiness without restrictions or instructions, thus fulfilling his/her desires and inclinations and realising his/her well-being.

3.2. Sensorimotor synergy

It is the child's ability to carry out various activities that require fine or large muscle movements and coordination between vision and movement, thus achieving a specific goal.

3.3. An autistic child

This is a child who has symptoms in three areas: Difficulties in social interaction, inability to communicate with others and the presence of stereotypical behaviours. The case was selected from the psycho-pedagogical centre who suffers from moderate degree autism and was previously diagnosed by the clinical psychologist.

3.4. Picture Matching Game

The game is designed to be both educational and enjoyable for children. It comprises a model of a bull and individual cards with the same image, which the child uses to match the pictures. This encourages the development of sensory-motor synergy and coordination between eye and hand movement.

4. THE THEORETICAL APPROACH

Play is one of the most significant needs that provide children with care, the discharge of emotions and enjoyment, which are fundamental aspects of childhood. It is also a fundamental activity that allows children to learn and develop skills, solve problems and integrate their mental development. Furthermore, play facilitates the child's socialisation, as it provides a means of communication with others. The game itself becomes the vehicle through which the child expresses his desires, inclinations and wants. In this way, the child is able to externalise what is in his being through his interaction with the game. [1, p. 2]

Montessori defines it as a major school in which the child grows and develops his physical, intellectual and even social strengths, thereby qualifying him in all respects for life. [2, p. 16] While Bruner defines it as a behaviour that allows the organism to discover and train new behavioural strategies [3, p. 22]. Play activity allows the child to integrate into his community and family, thus forming a normal physical and psychological structure through the acquisition of many skills in various aspects.

Play is considered an educational medium and an imperative for the child, as it is his favourite occupation in which he feels comfortable and catharsis, and a means of expressing his inclinations and desires and gaining positive energy. This is the case for a typical child, but if the child is diagnosed as autistic, then play for him is a mechanism to achieve social interaction and independence. The effectiveness of play in this context depends on the selection of the game and the choice of games according to the child's inclinations and desires in order to bring about change. The impact of play on the life of the autistic child depends on the selection of the game and the choice of the game according to the child's inclinations and desires. This is an important element in modifying undesirable behaviours and developing many aspects. The first of these is the sensory-sensory aspect. The sensory-motor aspect has a very important role, as autistic children tend to play games that stimulate their visual or auditory side. Furthermore, the use of colourful materials, such as coloured balls, cooking games, household utensils, soap bubbles and other games that facilitate motor development, is beneficial. These can be employed in a competitive setting, such as football, where two teams compete for a reward. Autistic children's play is often characterised by stereotypical behaviours, repetitive actions, a reluctance to change, an attachment to routine and a tendency to engage in circular and synthetic games with multiple colours, which are considered to be a form of visual stimulation. The following games are among those most frequently played by autistic children, according to their chronological age:

1- Ages from (less than 02 years old)

Children with autism at this stage prefer physical interaction, as they tend to play games that have a relationship with the sensory aspect, especially the visual sensory, such as playing with bubbles, as they are interested in the sight of these cases and playing with blocks, installing and re-disassembling them, which works to develop motor skills.

2- Pre-school ages: (02 to 04 years old)

At this stage, autistic children prefer toys that stimulate the imagination and rely on opening, closing, connecting and assembling, which works to develop their motor and visual skills, in addition to animal toys and identifying them based on their image or sound and linking them to their daily life, which helps the child to develop imaginative play and use musical instruments and prefer them even if they are used in the wrong way that helps them use the sensory-motor and emotional side. [4, p. 06]

3- Ages (05 years and above)

At this stage of the stages, we find that autistic children tend to advanced games, which include some games in which the child uses the keyboard and activities that give the child a kind of fun, amusement and excitement, including increasing his motivation to learn and discover, such as the swing - climbing with the accompaniment of another person and other games that are considered recreational in addition to developing some aspects of the autistic child

[4, p. 07]

The coordination and balance between all sensory and motor functions in the autistic child is considered one of the most crucial elements in the process of their learning and integration with their peers, whether they are within the oasis unit or the wider group to which they belong. The harmonious integration of the sensory and motor aspects allows the cognitive aspect of the child to flourish, as this is the most important aspect of their development. It is essential that children, whether they are neurotypical or on the autism spectrum, are able to discover, synthesise, analyse and learn. However, autistic children are unable to do this because they are unable to coordinate between the sensory and motor sides of their brains. This can be defined as the ability to carry out activities that require combining the skills of sight and movement for a unified goal between them. It is evident that autistic children are unable to coordinate between the sensory and motor sides. This can be defined as the ability to perform activities that require the combination of visual and motor skills for a unified goal between them. These two skills include fine and gross motor skills in addition to sensory-motor skills. Furthermore, these skills work to achieve a previously defined goal based on the harmony and coordination of muscle movement. [5, p. 124]

However, the autistic child is unable to do so due to the incomplete organisation of movement, which leads to many problems, including language, which is a means of communication, thinking and even perception. This makes it difficult for him to learn due to the difficulty in the previous tasks that lead to the hand being able to perform the task based on the eye seeing it. It is important to note that sight is essential in children's learning and is an engine for the other senses. [5, p. 124]

Although the symptoms of this type of difficulty that the autistic child experiences are similar to those of other typically developing children, the specific symptoms of this difficulty are as follows: difficulties in eating, sitting and sitting upright, standing quickly and walking, as well as difficulties holding objects in the hand and experiencing distraction and lack of concentration [5, p. 124]

As we observe that he is unable to perform tasks in a conventional manner, he is disinclined to engage in activities requiring the precise sensory input that is absent in his case, in addition to experiencing episodes of mental fugue and rapid boredom. This is due to his inability to carry out the activities offered to him in the adapted physical activity class, which depend on the The aforementioned symptoms are typically associated with autism and manifest as an inability to grasp objects with the index finger and thumb, as well as a decline in motor coordination. This is evidenced by the fact that the subject in question is unable to jump and run swiftly, which is attributed to the difficulty he experiences in using his limbs. [6, p. 02]

The use of play as a technique for the development of this aspect of the autistic child is considered necessary and inevitable in this category. It is used in the care of many cases as it helps the child to discharge emotional energy, vent and meet his needs. This enables the child to develop several aspects, including sensory-motor and fine movements, which in turn enables him to learn social learning. This ability to perform skills, whether mental, social or linguistic, is considered a mechanism to communicate with his peers and members of his community. Play is a means of communication between the child and the outside world, including his family, educators and peers. It is also a technique for the development of various aspects of his life, including the sensory-motor aspect that helps him to learn. The child achieves a goal previously set by the person in charge of this technique, whether a specialist, an art therapist or an educator. Furthermore, play is a means of recreation, ventilation and relaxation for the child. This is because the process of child development in all aspects, including the sensory-motor aspect, requires play as one of the parts of this process and an effective element in the development of these skills, which in turn leads to the development of cognitive and mental skills. In all aspects, including the sensory-motor aspect, play is a necessary component of this process and an effective element in developing these skills that lead to the development of cognitive and mental skills. A multitude of studies have confirmed and clarified the role of this type of technique in developing a multitude of skills, including sensory-motor synergy, which is the coordination between eye and hand movement.[8, p. 07]

5. APPLIED SIDE OF THE STUDY

As a preliminary step in the process, we contacted the specialists and educators working at the psycho-pedagogical centre, where a comprehensive observation was conducted of all cases exhibiting difficulty with sensory-motor synergy. These cases were diagnosed by the specialists as having an intermediate degree based on the Carrs Autism Scale and were diagnosed by the paediatrician. The cases in question have the potential to learn and communicate, and can improve their language and communication skills compared to children with a deep degree if their sensory-motor aspect is developed. One of these cases was selected for further work, with the aim of developing her sensory-motor synergy through play. This was achieved through the use of a picture matching game as a model.

The researchers conducted a review of the child's psychological and pedagogical report, which indicated that at the beginning of his enrollment in the centre, he exhibited difficulties in grasping objects and moving them in the appropriate manner, in addition to a tendency to hold the hand of the nanny or specialist in order to request assistance with the required activity. The mother stated during her interview with the specialist that her son had an issue with his hands, which prevented him from being able to move objects when instructed to do so.

5.1. Methodology of the study

In our study, we employed the clinical method, which is regarded as a scientific method comprising a set of procedures designed to elicit and analyse behaviours (Silamy 2003). This approach is based on the premise that the human being defines it by identifying all that is qualitative and individual in a person, within a very limited situation. Consequently, the clinical approach enables the collection of comprehensive data on autistic children and the formulation of an assessment of their psychological state, as well as the identification of the most significant issues they face, through the utilisation of the medical history of the case, based on the clinical interview with parents, specialists, educators and specialists. As an exemplification of this approach, the case study is the most appropriate methodology. This is because the psychologist works with each case individually, and each case is considered a unique entity that requires procedures tailored to its specific psychological abilities and mental level in order to achieve the previously set goals.

5.2. The case study

The case study represents a valuable opportunity for the specialist to gather as much information and data as possible, enabling him to gain a comprehensive understanding of the case through a dynamic, relational and even historical perspective. This approach allows the specialist to form a complete picture of the case, which in turn facilitates a comprehensive understanding of the case. [7, p. 95]

5.3. Research techniques

The current study will employ the following data collection techniques:

5.3.1. Clinical observation

Clinical observation represents one of the most crucial methods of data collection and psychological examination, enabling the gathering of a vast array of information about the subject under study from a multitude of perspectives. This approach is defined as a systematic observation and monitoring process, aimed at the detailed study of human behaviour and the accurate documentation of observed phenomena. [7, p. 99]. The case was observed in a variety of settings to assess sensory-motor harmony and coordination in different contexts, including within the unit and during adapted physical activity classes.

5.3.2. The clinical interview

The clinical interview was employed in this instance due to the characteristics of the autistic child, which is defined as a means of collecting as much data as possible by preparing and asking a set of questions with the aim of obtaining answers that help the researcher in his research. In this context, the examinee is permitted to answer the questions freely without deviating from the boundaries of the topic. In order to collect information about the case and provide a report that includes the psychological and behavioural aspects that characterised the case when the subject joined the centre, as well as all the questions that help in the study, the parents, the educator, the psychologist, the psychiatrist, the psychologist and the adapted physical activity teacher were interviewed. [7, p. 95]

5.4. The treatment program

The programme was developed by two researchers to enhance sensory-motor synergy in an autistic child. It comprises a series of activities designed to facilitate this aspect of development. The programme relies on a single game, the picture matching game, which has been shown to be effective in matching the card held by the child with the corresponding picture in the model, regardless of its format. Consequently, the child attains sensory-motor synergy. This was achieved through the implementation of a set of strategies and techniques, which were based on a significant set of different sources related to the subject matter. These sources included a review of the popular and theoretical heritage and psychological studies that targeted sensory-motor synergy for autistic children with autism. A number of these sources are as follows:

1- A Montessori-based mini-training programme to improve sensorimotor synergy in children with mild autism spectrum disorder. Prepared by Taghlit Salahuddin and Mokhiber Faiza (2018).
2- The effectiveness of a sensory integration-based training programme to develop fine motor skills in children with autism spectrum disorder. Prepared by Foueiris Rumaysa and Mahrezi Malika (2024).
3- The efficacy of a play-based programme in enhancing the psychomotor abilities of children with mild mental disabilities within the age range of 5 to 7 years. Prepared by Nahed Mounir Makari (2021), in addition to the book by Ibrahim Al-Hashimi, entitled A Guide to Games and Activities for Sensory-Motor Integration Training for Children with Autism and Mental Disabilities.

5.4.1. Timeframe

The programme was implemented over a period of two and a half months, with each of the 14 sessions lasting between 20 minutes and 45 minutes.

5.4.2. Method of applying the programme

In this study, the researchers employed the use of a picture matching game to facilitate the development of sensory-motor synergy. The number of sessions was set at 14, divided into three phases: exploratory, work, and completion.

5.4.3. Techniques employed in the programme

The following techniques were employed:

1- Reinforcement: This is a method that elicits a positive response from the child and eliminates the negative consequences of the behaviour, thereby encouraging the child to adopt the same behaviour in other situations. [9, p. 258]
2- Urging: This involves providing the child with adequate support to help them perform a certain behaviour, which increases their motivation to perform the same behaviour in the future.[9, p. 264]
3- Modelling: This entails the trainer or implementer of the programme performing the targeted behaviour or activity in front of the child, affording them the opportunity to observe it closely, before requesting that they repeat the behaviour and imitate it without assistance.[9, p. 261]

5.4.4. Programme content

The following table illustrates the content of the sessions and therapeutic exercises as outlined in the programme.

Number of sessions	Objectives	Therapeutic exercises
1	Case Interviewing and Case Selection Building a Therapeutic Relationship Pre-measurement	Welcome the situation Playing with the child in a free way Reviewing the case file and psychological report Pre-measurement
2	The mother of the case was interviewed in order to obtain preliminary information with the express permission to film and to introduce her to the programme and its objective. The next step was to build the therapeutic contract.	Provide a concise overview of the programme, including an estimation of its duration and the number of sessions. Additionally, gather pertinent information about the case history and the individual's knowledge of the issue they are suffering from.

Number of sessions	Objectives	Therapeutic exercises
3	The matching of the object presented to the child to its appropriate position.	The child is provided with the appropriate number of pieces and is instructed to arrange them in the correct manner without retaining any excess pieces.
4	Strengthen the match between the object and the place where it fits.	A child is presented with a considerable number of pieces, 12 in total, along with their own model, which contains 9 pieces. They are then asked to sort the matching pieces from the extra piece, which is to be set aside.
5	The objective is to achieve visual and motor coordination.	A model of a car is presented to the child, who is then asked to sort the pieces into two categories: those that fit into the model and those that do not.
6	Group play training.	The child should be engaged in a game with another child in which the game is presented and the child is asked to place a piece, with the expectation that their partner will place another piece, and so on until the model is finished.
7	The disruption of the child's routine is necessary in order to maintain the goal of sensory-motor development.	Provide the child with a plastic ball and allow them to engage in free play with it.
8	The disruption of the child's routine is necessary in order to maintain the goal of sensory-motor development.	The child is presented with a plastic ball and instructed to throw it to the researcher and catch it with their hands. If the ball does not fall, the child is
		praised for their ability to catch it without letting it fall. This reinforces the desired behaviour.
9	Development of fine motor skills.	Provide the child with the requisite number of pieces and a model, and instruct them to identify and segregate those pieces that are not included in the model.
10	Developing sensory-motor integration skills.	A model comprising a variety of automobiles is presented to the subject, who is then asked to place the piece in its appropriate position, thus matching it.
11	Achieve sensory-motor synergy.	A disassembled model of the human body is presented to the child, who is then asked to match the organs to their respective locations.
12	The objective is to develop sensory-motor synergy and coordination between eye and hand movements.	A model of the human body is presented to the child, who is then asked to match the part to its proper place. The child is also asked to point to the part with his finger in his body whenever he puts on a part.
13	A comprehensive overview of the topics discussed during the sessions.	A set of models comprising a variety of pieces is provided to the child, who is then asked to match them.
14	A meta-measurement and evaluation of the programme was conducted in relation to the case.	The child is presented with the game and asked to complete the activity independently with the nanny, with the researchers' involvement gradually

Number of sessions	Objectives	Therapeutic exercises
		decreasing.

5.4.5. Evaluation of the programme's effects

Following the completion of the programme, which comprised one game, the picture matching game, the extent of the development of sensory-motor synergy in the autistic child was determined based on their interaction with the game and the assistance of two researchers. This involved the implementation of a set of activities unified by a single objective, namely the development of sensory-motor synergy. The programme was applied by the nanny, the adapted physical activity teacher, and the mother to ascertain the nature and extent of the improvement observed in the child's activity. This was achieved by eliciting their opinions about the positive interaction of the child through the therapeutic sessions based on the picture matching game in the development of sensory-motor synergy before and after the programme was implemented.

6. PRESENT AND ANALYSE THE RESULTS OF THE STUDY

6.1. Introducing the case

F. Mustafa, a child aged six years and seven months, belongs to a family comprising four members (father, mother and a 10-year-old brother who resides with his maternal grandfather's family in another state due to the high costs associated with his upbringing and the father's inability to provide him with an adequate education). This information was provided by the child's mother during the interview. Mustafa has brown eyes and black hair, is neatly dressed and does not suffer from any diseases except for autism spectrum. At the beginning of the sessions, he was accompanied by his mother, and in the last sessions, he was accompanied by his father due to his mother's pregnancy. With regard to the economic situation of the family, it is weak, given that the mother is a stay-at-home mother and the father is a daily labourer.

Following a normal and desirable pregnancy, the mother's delivery of the case was normal with artificial feeding. Consequently, the mother was not exposed to any sudden injury or illness during that period. However, when the child reached the age of 10 months, the mother noticed A delay in crawling was observed, and the child was subsequently referred to a psychomotor therapist for massage. The mother then noted the absence of the child's first words and the lack of development in the language aspect, particularly in response to his name being called. Subsequently, the mother observed the absence of the infant's initial utterances and a lack of linguistic development, particularly in response to his name being called, as though he was unable to hear. Consequently, the parents sought the advice of a paediatrician, who conducted a hearing test at the age of four. This revealed no hearing impairment. However, the infant was subsequently referred to a psychiatrist for an autism diagnosis. He was subsequently referred to a psychologist, who was unable to make any discernible progress with him. The case was subsequently transferred to a private association for the purpose of providing care. During this period, the individual underwent a series of therapeutic sessions, during which the specialist attempted to modify his behaviour. However, due to the family's low economic situation, he was subsequently cut off from this support and was transferred to a psycho-pedagogical centre. It is notable that the mother demonstrated an acceptance of her son during this period. The mother demonstrated acceptance of her son by speaking in a comfortable and responsive manner and confirmed that she stands by his side at home and engages in activities with him when he returns in the evening. She stated, "I spend most of my time with him and I can help him as a stay-at-home woman." Furthermore, she indicates that following the birth period, she will refer the child to a psychologist and speech therapist in order to facilitate his development.

The case is characterised by complete calmness, a lack of screaming and the ability to sit in a chair. However, the subject displays some unusual behaviours, including twisting fingers, a lack of eye contact, a lack of response when called and a tendency to isolate himself from others. Furthermore, he does not perform any aggressive behaviours either towards himself or towards others. The case obtained an estimated score of 42 on the CARS scale, which indicates a moderate degree of autism symptoms. This was conducted by the clinical psychologist for the case when he joined the psycho-pedagogical centre in September 2023.

6.2. View the results obtained by the case through the treatment programme

Session number and time	Stage	Type of game	Session Objectives	Application Procedure	Results achieved											
					Types of help							The criterion		Result		
					EE	DD	Ro	Mo	P	PP	V	M	R	+	-	NR
1	The exploratory phase	A variety of games that engage the sensory-motor system.	Interviewing cases and selecting one case. Building a therapeutic relationship.	Welcome the situation Playing with the child in a free way Reviewing the case file and psychological report		*	*		*						*	
2		Initial data collection form from the mother of the case.	case to take the initial information with permission to film and introduce her to the programme	the programme Determine the approximate duration of the programme and the number of sessions Gather the necessary information	*	*										
3	Work Stage	A matching game with different images	Matching the object presented to the child to its proper position	Give the child only the right pieces and ask them to put them in the right place without having extra pieces left over			*	*	*		*				*	
4		Picture matching game with different images	Strengthen the match between the object and the place where	considerable number of pieces, 12 in total, along with their own model, which contains 9 pieces. They are then asked to sort the pieces into two categories:			*			*	*		*		*	
5		A matching game with car images	Achieve visual and motor coordination.	that contains only cars and give them pieces that fit with the model and others that have nothing to do with the model, asking them to sort the latter and			*			*					*	
6	Work Stage	Match game with different images	Group Play Training	child in a game where we present the game and ask the child to place a piece and wait for his/her mate to place			*	*			*		*	*		
7	Work Stage	Plastic ball	Breaking the child's routine while maintaining the same goal of sensory-motor	Give the child a plastic ball and let the child play with it freely				*			*			*		
8		Plastic ball	Breaking the child's routine while maintaining the same goal of sensory-motor	ball and ask him to throw it to the researcher and catch it with his hands, provided he does not drop the ball, and as he catches it without			*				*			*		
9	Work Stage	Match game with different ima7ges	Fine Motor Development	We give the child pieces and a model and ask them to sort the pieces that are not in the model and put			*						*	*		

Session number and time	Stage	Type of game	Session Objectives	Application Procedure	Results achieved											
					Types of help							The criterion		Result		
					EE	DD	Ro	Mo	P	PP	V	M	R	+	-	NR
10		A matching game with images of cars ranging in size from large to small	Sensorimotor integration skills development	We present a model with various cars and ask him to place the piece in its proper place, i.e. match it			*				*		*	*		
11	Work Stage	A game that matches images of the human body	Achieving Sensorimotor Synergy	We present the child with a disassembled model of the human body and ask them to match the organ to the			*					*	*	*		
12	The termination phase	A game that matches images to the human body	Developing sensory-motor synergy and eye-hand coordination	We present the child with a model of the human body and ask them to match the part to its proper place where we ask them whenever they put on a part, they point to it with a finger in their body.			*					*	*	*		
13	The termination phase	Various games about matching picture	Overview of what was discussed during the	We give the child a set of models with different pieces and ask them to match them			*					*	*	*		
14		Picture Matching Game Mounting Inflatables on a Plastic Pole	Conducting a meta-measurement and evaluating the programme	We offer the child the game and ask them to do the activity alone with the nanny without the intervention of the two researchers to gradually withdraw			*					*		*		

EE: Interviewing and Listening.
DD: Discussion and dialogue.
Ro: Reinforcement.
Mo: Modeling.
P: Total Physical Assistance.
PP: Partial Physical Assistance.
V: Verbal prompting.
M: Master the skill correctly.
R: Repeat multiple, valid attempts.
+: Correct response without assistance.
- : Incorrect response.
NR: No response.

7. ANALYSE AND DISCUSS RESULTS

The results presented in the preceding table clearly demonstrate that the hypothesis that the matching game has an impact on the development of sensory-motor synergy and the use of precise movements, as stated by the mother, is indeed accurate. In addition, the emergence of a set of other skills that help the subject in his daily life, such as visual communication, is also evident. By the conclusion of the sessions, the case exhibited an improvement in comparison to its previous state. This was evidenced by the emergence of social integration and seeking behaviours, as demonstrated by the case's tendency to enter among its colleagues, attempt to catch the ball from them and hug them when unsuccessful. These observations indicate that play, as a therapeutic technique, has a positive and effective impact. Additionally, the case's sensory-motor synergy, a result of the two researchers' continuous training and the intensification of their efforts, was observed. The educator and the family of the case were also involved in the process through daily follow-up and carrying out the duties required of them and following the advice and guidance provided by the two researchers. The latter employed a set of methods and techniques within the sessions, including reinforcement, which enabled them to ascertain the quality of the reinforcer that had an impact on the child. They also employed repeated prompting and modelling, which facilitated working with the child, especially when the activity was new for him. These techniques helped the case to respond to instructions and bring about positive change. The findings from the end of the sessions indicated that the matching game has a role in the development of sensory-motor synergy for the autistic child and that the game has a role in the development and improvement of sensory-motor synergy in the autistic child. This partially supports the main hypothesis.

8. CONCLUSION

In recent times, hospital institutions and psycho-pedagogical centres have observed a significant increase in the number of autistic children. This is a consequence of parents noticing a number of clinical symptoms in this group, including sensory-motor issues. In response, numerous researchers in the field of psychology have sought to develop therapeutic programmes with the objective of developing sensory-motor synergy for this group. This served as a source of inspiration for our study. A case study approach was employed in addition to the administration of an interview, observation, and the construction of a programme based on the picture matching game as a model for the development of this aspect. The results indicated that the programme had an effective impact on the development of sensory-motor synergy for this group through the picture matching game, which is considered one of the aspects of the daily life of the autistic child. The current study identified a number of recommendations, the most significant of which are:

The provision of toys and tools to facilitate psychomotor therapy for children with difficulties in this area is a mandatory requirement. It is also essential to ensure that staff are adequately trained and experienced in the field of sensory motor to provide assistance to children with special needs and autism. Furthermore, it is crucial to raise awareness among families of the importance of their participation in care programmes for their children:

- It is important to emphasise the role of the playroom in reducing some sensory response issues for children with autism.
- It is important to emphasise the effectiveness of sensory-motor activities in achieving social interaction for the school-integrated child.
- It is necessary to confirm the effectiveness of a play-based programme in reducing some manifestations of aggression in autistic children.

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